Thermal Distribution Collisions

e → i collisions

We are going to use the previously calculated results to find them for distribution functions instead of single particles.

We will ignore ion motion and attempt to average over the e^- distribution.

$$f_e(\vec{v}) = n_e\left(\frac{m_e}{2\pi T_e}\right)^{\frac{3}{2}} \exp\left[\frac{-m(\vec{v}-\vec{v}_d)^2}{2T}\right]$$

The momentum loss per electron is given by:

$$\nu_p = n_i \frac{q_e^2 q_i^2}{(4\pi\epsilon_0)^2} \frac{4\pi(m_e+m_i)}{m_i m_e^2 v^3} \ln\Lambda$$

For a single particle: $\frac{dp}{dt} = v\frac{dp}{dx} = v\frac{p\nu_p}{v} = p\nu_p = m_e\vec{v}\nu_p$

We have to put this value into the distribution: $-\frac{dp}{dt} = \int f_e(\vec{v})\,\nu_p(v)\,m_e\vec{v}\,d^3\vec{v}$

↳ The - sign is here because we have a rate of loss.

This is a tricky integral to evaluate. We can make the following assumptions to simplify it:

1.) Ignore variations of $\ln\Lambda$ with v. Set: $\Lambda = \lambda_D / b_{90}(v_t)$

↑ The inexact values required for $\ln\Lambda$ helps here.

2.) Assume that the drift velocity is small relative to the thermal velocity: $v_e \equiv \sqrt{T_e/m_e}$, we can then expand the drift velocity term to the first order. $\vec{u} \equiv \vec{v}/v_e$; $\vec{u}_d \equiv \vec{v}_d/v_e$

$$\therefore f_e = n_e \frac{1}{(2\pi)^{3/2} v_e^3} \exp\left[\frac{-1}{2}(\vec{u}-\vec{u}_d)^2\right] \simeq \frac{n_e}{(2\pi)^{3/2} v_e^3}(1+\vec{u}\cdot\vec{u}_d)\exp\left(\frac{-u^2}{2}\right) \equiv (1+u_x u_d) f_0 \rightarrow \text{unshifted Maxwellian}$$

↓ Taking drift along x dir^n.

Keep in mind that the thermal velocities and drift velocities are constant in this calculation.

Using this change of variables, we get the momentum loss rate p.u. volume:

$$-\frac{dp_x}{dt} = \int f_e \nu_p m_e u_x\, d^3u = \nu_p(v_t)\, m_e u_d \int \frac{u_x^2}{u^3} f_0\, d^3u$$

We can now evaluate this integral using the spherical symmetry of the unshifted Maxwellian.

$$\int \frac{u_x^2}{u^3} f_0\, d^3\vec{u} = \frac{1}{3}\int \frac{u_x^2+u_y^2+u_z^2}{u^3} f_0\, d^3\vec{u} = \frac{1}{3}\int \frac{u^2}{u^3} f_0\, d^3u = \frac{1}{3}\int \frac{u_e}{u} 4\pi f_0 u^2 du = \frac{2\pi}{3} v_e \int_0^\infty f_e\, 2u\, du$$

$$= \frac{2}{3(2\pi)^{1/2}} n_e$$

Thus we get the "Maxwell-averaged momentum-loss frequency":

$$\frac{-1}{p}\frac{dp}{dt} \equiv \bar{\nu}_{ei} = \frac{2}{3(2\pi)^{1/2}}\nu_p(v_t)$$

In more explicit terms: (Assuming m_e/m_i to be very small)

$$\bar{\nu}_{ei} = \frac{2}{3(2\pi)^{1/2}} n_i \left(\frac{ze^2}{4\pi\epsilon_0}\right)^2 \frac{4\pi}{m_e^{1/2}T_e^{3/2}} \ln\Lambda_e$$ → Standard form of the electron collision frequency.

* ion to electron collisions are the same.

i → i Collisions

The difficulty in ion → ion collisions is that both parties are moving. We can do this by accounting for the relative velocity. This way, the average rate of transfer p.u.volume:

$$-\frac{d\vec{p}}{dt} = \iint \vec{v}_r \frac{m_1 m_2}{m_1+m_2} v_r 4\pi b_{90}^2 \ln\Lambda\, f_1 f_2\, d^3v_1\, d^3v_2$$

↳ v_r is the relative velocity $(\vec{v}_1 - \vec{v}_2)$

For our sanity, we can shift the integral into being expressed by the relative velocity and average velocity.

$$\vec{v}_r \equiv \vec{v}_1 - \vec{v}_2 \quad ; \quad \vec{v} \equiv \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$$

f_1 and f_2 are shifted maxwellians in the cm frame: $f_j = n_j\left(\frac{m_j}{2\pi T}\right)^{3/2}\exp\left[-\frac{m_j(\vec{v}_j - \vec{v}_{dj})^2}{2T}\right] \quad (j=1,2)$

Here, $m_1\vec{v}_{d1} + m_2\vec{v}_{d2} = 0$

This gives us the product in cm coords:

$$f_1 f_2 = n_1 n_2 \left(\frac{m}{2\pi T}\right)^{3/2}\left(\frac{m_r}{2\pi T}\right)^{3/2}\exp\left[\frac{-mv^2}{2T} - \frac{m_r v_r^2}{2T}\right]\left\{1 + \frac{m_r}{T}\vec{v}_d\cdot\vec{v}_r\right\}$$

We can plug all this and the integral becomes similar to that for e → i collisions.

$$-\frac{dp}{dt} = \frac{2}{3(2\pi)^{1/2}}\left(\frac{q_1q_2}{4\pi\epsilon_0}\right)^2\frac{4\pi}{m_r^2 v_{rt}^3}\ln\Lambda_t\, n_1 n_2 m_r v_d$$

↳ $v_{rt} \equiv \sqrt{T/m_r}$

This is the general momentum exchange rate for two maxwellians drifting at small relative velocity v_d.

To get the collision frequency, we just have to decide which target is stationary. If 2 is target:

$$\bar{\nu}_{12} = \frac{1}{n_1 m_1 v_d}\frac{dp}{dt} = \frac{2}{3(2\pi)^{1/2}} n_2\left(\frac{q_1q_2}{4\pi\epsilon_0}\right)^2\frac{4\pi}{m_r v_{rt}^3}\frac{\ln\Lambda_t}{m_1}$$

For equal mass ions: $v_{rt} = \sqrt{\frac{T}{m_r}} = \sqrt{\frac{2T}{m_i}} \longrightarrow \bar{\nu}_{ii} = \frac{1}{3\pi^{1/2}} n_i\left(\frac{q_1q_2}{4\pi\epsilon_0}\right)^2\frac{4\pi}{m_i^{1/2}T_i^{3/2}}\ln\Lambda$

Thus, the ion-ion collision frequency is $1/\sqrt{2}$ times the electron ion collision frequency.

* $e \to e$ collisions can be described by subbing $i \to e$.

A problem we encounter while describing $e \to e$ collisions is how two maxwellians can be distinguished, as all electrons are virtually indistinguishable.